

V3_30 CFD Findings

How the new CAD and the CFD thrust result were put into the simulator, what was flown, and why the conclusion is that V3_30 does not have enough pitch control with the CFD thrust.

Conclusion. The CFD (run on foils of 75 / 65 / 50°) shows each side's jets push mostly forward: 56 N forward and 46 N up per group, 67.5 % of the fan thrust. To cancel that forward push, V3_30 has to hover at **42.7° nose-up** instead of 27°. At that angle only **1.5 N·m** of pitch moment is left in reserve, against 7.4 N·m before. In SITL it tips nose-down at lift-off, or gets airborne with its motors at the limit most of the time and crashes. Gains cannot fix this; the fan and CG layout has to change.

1. The workflow

1 Read the new CAD's top-level folders.

Listed the 53 top-level component folders of `SMALL_SCALE_V3_30.step` with their volume and centre. Only the weight labels in the folder names were used (for example `ESC_M200g` = 200 g), spread over each folder's parts by volume. Folders without a label count as 0 g. Result: 14.40 kg, centre of gravity 11 cm behind the boom centre.

2 Compared it with the previous CAD (v34).

Same reading on the old STEP, then matched parts by position. Result: four batteries moved 195 mm aft, avionics moved 47 mm, inner foils redrawn, fans unchanged.

3 Measured the geometry the simulator needs.

Ray-traced each foil to get the jet exit angle and trailing edge; found the lowest point of each foot. Result: foils 49.5 / 58.2 / 49.6° (inner / middle / outer); rests on two rear feet and the front foot at -17.6°.

4 Built the simulator model and checked it standing still.

Created the V3_30 airframe; checked the rest angle, thrust-to-weight and the hover mix. Result: rests at -18.2°, hovers at 27.15°, thrust-to-weight 2.20, every fan 41-59 % in hover.

5 SITL flights and tuning (before CFD).

Flew the rotate-and-hover batch headless over 3 seeds, then grids of pitch gains. Result: pitch-rate P 0.75 → 0.9, I 0.1 → 0.05; drift 13.4 → 8.4 m, touchdown 0.28 → 0.20 m/s. flies

6 Live flights in the GUI.

Flew the tuned model live; much worse than headless, which exposed a saved magnetic declination of 5° in the app's PX4. Result: fixed in the SITL export (declination 0); heading drift 17° → 2.9°.

7 Put the CFD thrust into the model.

Converted the CFD group force into a per-fan force: each foil fan 24.3 N at 39.6° above the duct axis (67.5 % of 36 N). Result: the CFD airframe.

8 Re-trimmed the hover.

The hover-angle search stopped at 40°, where no fan mix works; widened it to 80°. Result: hover at 42.7° in a window only a few degrees wide, hover throttle 0.65, thrust-to-weight 1.80.

9 SITL calibration flights with the CFD model.

First: PX4 refused to arm ("Attitude failure"): parked, it sees itself 60° nose-down. Raised `FD_FAIL_P` to 70°. Then: the nose fans rotate it to 42.7°, but at lift-off it tips onto its nose. With faster lift-off ramps it gets airborne, motors saturated 60-70 % of the time, and crashes.

does not hover

10 Found the cause with a static margin check.

Computed how much roll and pitch moment is left at hover before a fan hits 0 or 100 %, for the old model, the CFD model and what-ifs. Result: pitch reserve 7.4 → 1.5 N·m. Better foils (80-90 % efficient, or steeper jets) only reach 2-3 N·m.

11 Checked the foil set.

The CFD was for 75 / 65 / 50° foils, not V3_30's own. Moved the force points to the nearest steep-foil CAD (v34). Result: 43.0° hover, 1.42 N·m: same conclusion.

2. Files: inputs, outputs and what was done with them

Paths are relative to the repository. `results/` is not tracked by git.

CAD reading and geometry

FILE	INPUT	OUTPUT	WHAT I DID WITH IT
<code>airframes/cad/SMALL_SCALE_V3_30.step</code>	Your CAD (copied from UtopiaLabs/)	(source)	Read by every script below
<code>results/v3_30/step_tree.py</code>	The STEP	<code>tree.json</code> : every folder's name, number of parts, volume, centre	Picked the 53 top-level folders and their weight labels
same script, in <code>results/v3_30/v34/</code>	The old v34 STEP	<code>v34/tree.json</code>	Compared with V3_30 to see what moved
<code>results/v3_30/bboxes.py</code>	STEP, folder names	Bounding box per folder	Located the feet and the unnamed, unlabelled solid at the right rear
<code>results/v3_30/foot_points.py</code>	STEP, foot folders	Lowest contact point of each foot part	Leg positions in the model; parked angle -17.6°
<code>results/v3_30/foil_rays_step.py</code>	STEP (and v34 STEP to check the method)	<code>foil_exit_deg.json</code> , <code>foil_profiles.json</code>	Exit angles 49.5 / 58.2 / 49.6° and trailing edges, which set where each foil fan's force acts

Simulator models

FILE	INPUT	OUTPUT	WHAT I DID WITH IT
<code>scripts/atlas_v3_30_from_step.py</code>	STEP, <code>tree</code> , exit angles and profiles; gains, fan model and H-FLOW mount from <code>atlas_v3_v34_foils_50_65_50_tuned.json</code> ; fan axes from <code>atlas_v3_v34.json</code>	<code>airframes/atlas_v3_30.json</code> : 14.40 kg, rests -18.2°, hovers 27.15°	The model for all pre-CFD flights
<code>airframes/atlas_v3_30_tuned.json</code>	<code>atlas_v3_30.json</code> + the tuned pitch gains	Tuned model	Base for the CFD model and the live flights
<code>scripts/atlas_v3_30_cfd.py</code>	CFD group force (-56.12, 46.48, 0.73) N, 36 N per fan, foils 75 / 65 / 50°, <code>atlas_v3_30_tuned.json</code>	<code>airframes/atlas_v3_30_cfd.json</code> : foil fans 24.3 N at 39.6°, hover 42.7°, throttle 0.65, <code>FD_FAIL_P</code> 70	The model for the CFD flights and margin check

Flights and analysis

FILE	INPUT	OUTPUT	WHAT I DID WITH IT
<code>scenarios/2026-10-01_1511_v3_30_cad_rotate_hover.json</code>	(the batch)	Rotate from park, lift off, 60 s Stabilized hover, land	The flight used for tuning and the CFD calibration
<code>scenarios/2026-10-01_1523_v3_30_cad_rotate_poshold_hflow.json</code>	(the batch)	Same with a 60 s Position hold on the H-FLOW	Checked the tuned gains hold position (0.12-0.19 m)
<code>results/v3_30/fly.py</code>	Airframe, batch, gain variants, seeds	Summary per flight (errors, drift, touchdown, saturation) + time series	Ran every SITL flight below, several in parallel
<code>results/v3_30/diag.py</code>	A flight's time series	Steady offset vs wobble, estimate vs truth, speed, fan use	Found the steady nose-down offset; chose what to tune
<code>baseline.json</code> , <code>step1_integrator.json</code> ... <code>step5_refine.json</code> , <code>step4_candidates.json</code>	V3_30 with gain variants	Flight summaries	Picked pitch-rate P 0.9, I 0.05
<code>poshold.json</code>	Tuned vs old gains, Position hold	Flight summaries	Confirmed the tuned gains
<code>results/tuning/live893647940</code> , <code>live893903707</code>	Live GUI flights	Flight summaries + time series	Found and fixed the saved declination
<code>cfd_baseline.json</code>	CFD model, 3 seeds	Arming refused, then no lift-off	Raised <code>FD_FAIL_P</code> ; found the nose-down tip at lift-off

FILE	INPUT	OUTPUT	WHAT I DID WITH IT
<code>cfid_ramp1.0.json</code> , <code>cfid_ramp0.5.json</code>	CFD model, faster lift-off	Airborne, saturated 60-70 %, crashes	Showed it is an authority problem, not a lift-off problem
<code>results/v3_30/margins.py</code>	Old, CFD and what-if models	Hover angle, fan range, spare roll and pitch moment	The table below; the main evidence for the conclusion

Code changed in the simulator

FILE	CHANGE	WHY
<code>geometry/airframe.py</code>	Hover-angle search up to 80° (was 40°)	The CFD hover is at 42.7°; no other airframe's trim changes
<code>geometry/airframe.py</code>	SITL export sets magnetic declination to 0	The simulated compass has 0°; PX4 had a saved 5°
<code>sensors/models.py</code> , <code>sim/scenario.py</code>	Comment and error message	Match the two changes above

3. The key numbers

CASE	HOVER PITCH	FAN RANGE AT HOVER	SPARE ROLL (N·M)	SPARE PITCH (N·M)
V3_30, old jet model	27.1°	41-59 %	5.5	7.4
V3_30, CFD thrust	42.7°	24-86 %	5.9	1.5
CFD thrust, steep-foil force points	43.0°	23-87 %	5.6	1.4
What-if: foils 80 % efficient	42.6°	24-75 %	13.5	2.9
What-if: foils 90 % efficient	42.6°	25-67 %	19.0	3.2
What-if: jets 50° up, 67.5 %	29.8°	37-76 %	9.6	2.5
What-if: jets 55° up, 67.5 %	24.6°	43-71 %	11.3	2.1

What is assumed

The CFD gives the force of each side's group, not how it splits between the three fans or where it acts, so all three fans in a group get the same direction and their force stays on their own trailing-edge line. The 0.73 N spanwise part (1 %) is ignored because its sign was not given. Fan thrust (36 N), reaction torque and spool time are still the unmeasured defaults.

Reproduce: `.venv/bin/python scripts/atlas_v3_30_from_step.py` , `scripts/atlas_v3_30_cfd.py` , then `results/v3_30/margins.py` . Notes in `brain/experiments/2026-10-01-v3-30-step-import-and-tune.md` .